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Design and Analysis of Oversampling ADC

A Minor Project Report (XX367P)

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Department of Electronics and Communication Engineering

CERTIFICATE

Certified that the minor project (XX367P) work titled ***Design and Analysis of Oversampling ADC*** is carried out by **P Narashimaraja** (**1RV22EC005**), **Nithin M** (**1RV22EC006**), **Aditya** (**1RV22EE007**) and **Bala N** (**1RV22EC007**) who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in **Electronics and Communication Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2024-25. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the minor project report deposited in the departmental library. The minor project report has been approved as it satisfies the academic requirements in respect of minor project work prescribed by the institution for the said degree.

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DECLARATION

We, **P Narashimaraja**, **Nithin M**, **Aditya** and **Bala N** students of sixth semester B.E., Department of Electronics and Communication Engineering, RV College of Engineering, Bengaluru, hereby declare that the minor project titled '**Design and Analysis of Oversampling ADC**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in **Electronics and Communication Engineering** during the year 2024-25.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru

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ABSTRACT

The healthcare sector generates vast quantities of unstructured clinical data through documents such as referral letters and discharge summaries. Manual processing of these records is time-consuming and error-prone, creating significant bottlenecks in clinical workflows. This project addressed the challenge of automated medical document understanding and Systematized Nomenclature of Medicine — Clinical Terms (SNOMED CT) coding within a template-free intelligent extraction framework.

The domain of clinical Natural Language Processing (NLP) has expanded rapidly with the adoption of Electronic Health Records (EHR) and the growing volume of clinical documents requiring structured coding. However, many existing systems rely on rigid, template-dependent approaches that fail to adapt to diverse document layouts commonly encountered in real-world National Health Service (NHS) environments. This limitation often results in incomplete or inaccurate code mappings.

The proposed system, developed as an Intelligent Clinical Document Processing System (ICDPS), aimed to: (i) create an intelligent ingestion module for PDF and image-based clinical documents; (ii) develop a non-template-based extraction engine for diverse layouts; (iii) automate clinical summarization and categorization into predefined medical fields; and (iv) normalize free-text clinical concepts to SNOMED CT codes.

The methodology followed an iterative Software Development Life Cycle (SDLC) consisting of requirements analysis, design, implementation, testing, and deployment. OCR pipelines handled image-based inputs, while direct extraction processed PDFs. Transformer-based NLP models were employed for Named Entity Recognition (NER) and classification of entities such as Diagnosis, Medication, Procedure, and Allergy. A confidence-ranked SNOMED CT suggestion interface assisted users in selecting suitable codes.

The system achieved clinical entity extraction accuracy exceeding 88% and SNOMED CT mapping precision of 85% on the test dataset, satisfying project requirements. The resulting system demonstrated potential to reduce administrative burden, accelerate structured record generation, and improve the quality of medical coding in clinical environments.

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INTRODUCTION

Clinical document processing and automated medical coding have emerged as important research areas due to the rapid growth of unstructured healthcare data and the increasing need for efficient information management. Understanding the challenges associated with extracting meaningful information from heterogeneous clinical documents is essential for developing intelligent healthcare solutions capable of supporting clinical workflows. The proposed Intelligent Clinical Document Processing System (ICDPS) with SNOMED CT Coding addresses these challenges through the integration of document understanding, natural language processing, and medical terminology mapping techniques. The chapter presents the background and significance of the problem domain, discusses the project objectives and scope, examines existing research through a literature review, identifies research gaps, and outlines the methodology and organization of the overall work.

1.1 Introduction

Clinical document processing represents one of the most labour-intensive and administratively demanding activities in modern healthcare systems. Hospitals and healthcare institutions generate thousands of documents daily, including referral letters, discharge summaries, outpatient records, diagnostic reports, and clinical notes. These documents contain valuable patient information necessary for diagnosis, treatment planning, and long-term healthcare management. However, much of this information exists in unstructured free-text form, making efficient processing and analysis difficult.

Historically, clinical coding and information extraction have been performed manually by trained healthcare professionals and clinical coders. While manual approaches provide flexibility, they are time-consuming, expensive, and susceptible to inconsistencies arising from subjective interpretation and human error. As healthcare systems increasingly adopt Electronic Health Records (EHRs), the volume of digital clinical data has expanded significantly, creating an urgent need for intelligent automation methods capable of processing information efficiently.

Recent developments in Artificial Intelligence (AI), Natural Language Processing (NLP), and deep learning technologies have created opportunities for transforming clinical document processing workflows. Advanced transformer-based architectures can understand contextual relationships in medical text and extract clinically meaningful information from heterogeneous document formats.

The proposed Intelligent Clinical Document Processing System (ICDPS) with SNOMED CT Coding aims to address these challenges through a template-independent framework capable of processing PDF and image-based clinical documents. The system integrates Optical Character Recognition (OCR), Named Entity Recognition (NER), and ontology-based clinical coding to convert unstructured clinical narratives into structured and meaningful information representations suitable for healthcare applications.

1.1.1 Terminology and Definitions

The development of the Intelligent Clinical Document Processing System (ICDPS) involves several concepts and technical terms from healthcare informatics, Natural Language Processing (NLP), Artificial Intelligence (AI), and clinical coding systems. Understanding these terms is essential for interpreting the design and implementation aspects of the proposed system. The important terminology used throughout this report is presented below.

- **Clinical Document:** A healthcare-related document containing patient information, medical history, diagnoses, prescriptions, laboratory reports, referrals, discharge summaries, and clinical observations.
- **Natural Language Processing (NLP):** A branch of Artificial Intelligence concerned with enabling computers to understand, process, analyze, and generate human language.
- **Optical Character Recognition (OCR):** A technology used for converting text embedded within scanned documents or images into machine-readable textual information.
- **Named Entity Recognition (NER):** A Natural Language Processing task used to identify and classify specific entities from text into predefined categories such as diseases, medications, procedures, symptoms, and patient details.
- **SNOMED CT:** Systematized Nomenclature of Medicine – Clinical Terms, a comprehensive clinical terminology system used to represent medical concepts in a stan-

dardized manner.

- **Electronic Health Record (EHR):** A digital version of a patient's medical history that includes clinical data collected and maintained across healthcare systems.
- **Transformer Model:** A deep learning architecture that uses attention mechanisms to capture contextual relationships within sequential data and is widely used in modern NLP systems.
- **Intelligent Clinical Document Processing System (ICDPS):** The proposed framework designed to automate extraction of clinical information from heterogeneous healthcare documents and map extracted information to standardized SNOMED CT codes.

1.2 Overview of the Project Topic

The domain of intelligent clinical document processing sits at the intersection of healthcare informatics, natural language processing, and machine learning. The ability to automatically extract structured information from clinical narratives has profound implications for patient safety, clinical efficiency, and healthcare analytics. This section provides a broad overview of the domain from global, societal, and problem-specific perspectives.

1.2.1 Global Scenario

Globally, healthcare systems are undergoing a digital transformation driven by EHR adoption, telemedicine, and data-driven clinical decision support. According to reports from the World Health Organization (WHO) and major health informatics bodies, the volume of clinical data generated per patient encounter has grown exponentially over the past decade. In the United Kingdom, the NHS processes over one million patient interactions every 36 hours, generating vast quantities of unstructured clinical text. Similar volumes are observed in the United States, where the Centers for Medicare and Medicaid Services (CMS) mandates structured clinical documentation for reimbursement purposes.

Research in clinical NLP has accelerated significantly since the release of large pretrained language models. The i2b2 (Informatics for Integrating Biology and the Bedside) shared task competitions have benchmarked extraction performance across diverse clinical entity types. Recent results from these competitions show that transformer-based systems consistently outperform traditional rule-based and feature-engineered approaches. The adoption of models such as BioBERT [1], ClinicalBERT [2], and BioMedBERT has set new state-of-the-art benchmarks for clinical NER and relation extraction tasks.

Key organizations including NHS Digital, HL7 International, and SNOMED International are actively developing frameworks for clinical coding automation. The global market for AI in healthcare was valued at approximately USD 15.4 billion in 2022 and is projected to reach USD 187.7 billion by 2030, with clinical NLP representing a significant segment of this growth.

1.2.2 Societal Relevance

The societal impact of automated clinical document processing is multi-dimensional.

For patients, accurate and timely clinical coding ensures that diagnoses and treatments are correctly recorded, which directly influences care continuity, insurance claims, and epidemiological data accuracy. Errors in clinical coding have been linked to incorrect treatment pathways, insurance disputes, and flawed public health statistics.

For healthcare professionals, automated extraction reduces the administrative burden associated with manual coding and documentation review. Clinicians in the NHS spend an estimated 15–20% of their working time on administrative tasks, including documentation. By automating routine extraction and coding, the proposed system frees clinicians to focus on patient care.

From a public health perspective, accurately coded clinical data is essential for disease surveillance, resource planning, and epidemiological research. Standardized SNOMED CT coding enables interoperability across national and international health information networks, facilitating large-scale clinical research and policy formulation.

1.2.3 Problem Area

Despite the evident need for automated clinical coding, existing commercial and research systems exhibit significant limitations. The majority of deployed clinical NLP pipelines are template-dependent: they rely on predefined document structures and keyword lists, making them brittle when confronted with the diversity of real-world clinical documents authored by different institutions, specialties, and individual clinicians.

A secondary problem lies in the complexity and scale of SNOMED CT itself, which contains over 350,000 active concepts and 1.5 million relationships. Mapping free-text clinical entities to the most semantically appropriate SNOMED CT concept requires nuanced contextual understanding that traditional rule-based systems cannot provide. Current mapping tools often suggest multiple potentially correct codes without adequate confidence ranking, requiring significant manual intervention to select the most appropriate code.

Furthermore, existing solutions typically operate on digitally native text, failing to handle image-based documents such as scanned referral letters — a common format in many clinical settings where legacy paper-based workflows persist. The consequence is a significant gap between the theoretical capability of NLP technology and its practical utility in diverse NHS environments.

1.3 Specific Details of the Project Topic

The proposed Intelligent Clinical Document Processing System (ICDPS) integrates several technical components to address the identified limitations. The system is designed around a modular pipeline architecture comprising four primary layers: document ingestion, information extraction, clinical summarization and categorization, and SNOMED CT mapping.

The document ingestion layer accepts PDFs and image-based documents. For PDFs containing selectable text, a direct text extraction module using the PyMuPDF library is employed. For image-based documents — including scanned letters and photographs — an OCR engine powered by Tesseract 5.0 with LSTM models performs character recognition. Both pathways produce a normalized UTF-8 text representation of the document, which is passed to the extraction layer.

The information extraction layer is the core of the ICDPS. It employs a fine-tuned BioBERT model for clinical NER, trained on the i2b2 2010 and 2012 clinical NLP datasets augmented with NHS-specific annotations. The NER model identifies and classifies clinical entities into the following categories like Diagnosis, Medication, Dosage, Procedure, Anatomy, Allergy, Lab Result and Temporal Expression. A secondary classification module assigns each extracted entity to a role-specific action category (e.g., "For Pharmacist Action", "For Clinician Review") to support multi-role clinical workflows.

The summarization module employs an extractive summarization approach using sentence importance scoring based on entity density and clinical relevance metrics. The output is a structured summary containing the most clinically significant sentences organized by entity category.

The SNOMED CT mapping module uses a two-stage approach. In the first stage, candidate SNOMED CT concepts are retrieved using a vector similarity search against a pre-built SNOMED CT embedding index constructed using FastText embeddings of concept descriptions. In the second stage, a reranking model assigns confidence scores to the retrieved candidates based on contextual matching with the source document.

1.4 Purpose and Motivation

The primary purpose of this project is to reduce the administrative burden on NHS clinical staff by providing an intelligent, automated solution for extracting and coding information from clinical documents. The system is motivated by direct observation of inefficiencies in clinical coding workflows, where trained coders manually process hundreds of documents per day with limited automation support.

A secondary motivation is the opportunity to improve the quality and consistency of SNOMED CT coding across diverse clinical environments. By providing confidence-ranked code suggestions derived from a semantically aware NLP pipeline, the system reduces the cognitive load on coders and decreases the likelihood of incorrect or incomplete coding.

The project is also motivated by the growing availability of pre-trained biomedical language models and clinical NLP benchmarks, which have made it feasible to build high-accuracy extraction systems without requiring massive proprietary training datasets. The combination of open-source tools, publicly available biomedical corpora, and modern transformer architectures enables the development of a production-grade system within an academic project timeline.

1.5 Problem Statement

Current clinical document processing systems lack a standardized, template-free mechanism for automatically extracting clinical intent and mapping it to medical terminologies such as SNOMED CT without extensive manual intervention. Existing systems suffer from template dependency — the inability to generalize across heterogeneous document structures — and fail to accommodate image-based documents that predominate in legacy clinical workflows. The consequence is a persistent gap between the volume of clinical text generated and the volume that can be efficiently structured and coded.

This project addresses the following specific problem: given a heterogeneous collection of clinical documents in PDF or image format, automatically identify and classify clinical entities, generate structured summaries organized by clinical role, and suggest ranked SNOMED CT codes for each identified entity with confidence scores exceeding 85%, without reliance on fixed document templates.

1.6 Objectives

The objectives of the project are:

1. To develop an intelligent document ingestion module capable of processing clinical documents in both PDF and image formats, using OCR and direct text extraction techniques.

2. To implement a non-template-based clinical information extraction engine using fine-tuned transformer models (BioBERT) for NER across eight clinical entity categories.

3. To automate the generation of structured clinical summaries and categorize extracted entities into role-specific action groups for Pharmacist and Clinician review.

4. To develop a SNOMED CT mapping module that retrieves and ranks candidate codes using semantic vector similarity and a confidence-based reranker, achieving mapping precision above 85%.

5. To validate the complete end-to-end system on a curated test corpus of NHS-representative clinical documents and report standard NLP evaluation metrics.

1.7 Scope and Relevance

The scope of this project encompasses the design, development, and validation of the Intelligent Clinical Document Processing System (ICDPS) for processing English-language clinical documents using Artificial Intelligence and Natural Language Processing techniques. The proposed system focuses on transforming unstructured clinical information into structured and standardized outputs to improve the accessibility and usability of healthcare data.

The system supports the ingestion of various document formats including typed PDF documents, scanned image-based documents such as JPEG, PNG, and TIFF files, as well as multi-page clinical reports. It incorporates OCR and layout-aware document understanding techniques to extract text from scanned records and medical documents. The extracted information is further processed using NLP methods for clinical entity extraction, semantic understanding, and summarization. The system focuses on predefined clinical entity categories and maps identified medical concepts to standardized SNOMED CT codes to ensure semantic interoperability. The processed information is then converted into structured machine-readable formats such as JSON. The proposed system also integrates a human-in-the-loop validation mechanism that enables users to review and verify extracted entities and generated code suggestions before producing the final output.

Certain functionalities are considered outside the scope of this project. The system does not perform direct diagnosis or clinical decision-making and is not intended to function as a clinical decision support system. Real-time patient monitoring, integration with live Electronic Health Record systems, and processing of handwritten clinical notes beyond printed-text OCR capabilities are not addressed within this work. Support for non-English clinical documents and highly specialized domain-specific edge cases requiring expert medical interpretation are also beyond the scope of the current implementation. Furthermore, the project is limited to a prototype-level implementation in a controlled environment and does not include large-scale deployment in production healthcare settings. The system currently operates in an offline batch-processing mode, while real-time API deployment and large-scale healthcare integration are considered future extensions.

The implementation assumes that input clinical documents are of reasonable quality, including readable scanned images and properly formatted reports. It also assumes the availability of cloud-based services and APIs required for OCR, NLP processing, and ontology mapping tasks, along with access to standard medical ontologies such as SNOMED CT. Certain constraints may affect system performance, including computational requirements of transformer-based models, dependency on external services, OCR limitations for low-quality images, and variations in document structures and formatting styles.

The project has significant relevance across industrial, research, and societal domains. From an industrial perspective, the system addresses challenges associated with managing large volumes of unstructured clinical data by reducing manual workload, improving interoperability with healthcare information systems, and lowering operational costs. From a research perspective, the work contributes to ongoing advancements in AI-driven healthcare solutions by integrating document understanding, OCR, NLP, semantic search, and ontology mapping techniques into a unified processing framework. From a societal perspective, improving the accuracy and efficiency of clinical document processing can support better patient care, reduce the risk of medical errors, and enhance the reliability and effectiveness of healthcare services. The proposed system can benefit hospitals, healthcare institutions, clinicians, healthcare IT platforms, medical researchers, health-tech organizations, and digital healthcare providers.

1.8 Literature Review

A comprehensive review of existing research was conducted to understand the state of the art in clinical NLP, document processing, and SNOMED CT coding. The following subsections present a structured survey of relevant peer-reviewed publications and identify the research gaps that motivated the proposed system.

1.8.1 Literature Survey

The literature survey presented in Table 1.1 provides a comprehensive review of existing research works and technologies relevant to the proposed Intelligent Clinical Document Processing System (ICDPS). The survey includes studies related to biomedical language models, clinical natural language processing, named entity recognition, ontology mapping, document understanding, OCR techniques, information retrieval, relation extraction, de-identification methods, and clinical AI systems. Each research work has been analyzed based on its methodology, key findings, advantages, limitations, and relevance to the proposed project. The analysis of these studies provides an understanding of current advancements, identifies existing research gaps, and supports the selection of suitable techniques and models for developing the proposed system. The findings obtained from the literature survey also serve as a foundation for designing an integrated framework capable of efficiently processing, extracting, and standardizing information from unstructured clinical documents.

Table 1.1: Literature Survey of Representative Research Works

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[1]	Lee J. et al.	BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining	2020	Pre-trained BERT on biomedical corpora (PubMed and PMC) and fine-tuned on biomedical NLP tasks	Domain-specific pretraining significantly improved biomedical NER and relation extraction performance	High contextual understanding of biomedical terms; state-of-the-art performance	Requires large computational resources and extensive training data	Forms the primary NER backbone for the proposed ICDPS system
[2]	Alsentzer E. et al.	Publicly Available Clinical BERT Embeddings	2019	Fine-tuned BERT on MIMIC-III clinical notes	Clinical-specific training improved understanding of healthcare text	Better contextual representation for clinical language	Limited by availability of domain-specific clinical datasets	Used as an alternative NER backbone during model comparison
[3]	Spasic I., Nenadic G.	Clinical Text Data in Machine Learning: Systematic Review	2020	Systematic review of 67 clinical text mining studies	Identified data scarcity and annotation challenges	Comprehensive review of clinical NLP limitations	No experimental implementation	Guided transfer learning and data augmentation approaches
[4]	Stubbs A., Uzuner O.	Annotating Longitudinal Clinical Narratives for De-identification	2015	Clinical document annotation for PHI detection	Established benchmark dataset for clinical NER evaluation	Standardized annotation framework	Focused primarily on de-identification tasks	Influenced entity categorization in the proposed system
[5]	Suominen H. et al.	Overview of the ShARe/CLEF eHealth Evaluation Lab 2013	2013	Shared clinical NER task evaluation	Best systems achieved high disorder recognition accuracy	Provides benchmark datasets	Limited dataset diversity	Used as baseline for model performance comparison
[6]	Savova G.K. et al.	Mayo Clinical Text Analysis and Knowledge Extraction System (cTAKES)	2010	Rule-based and NLP-based clinical information extraction	Effective extraction of clinical concepts	Open-source and modular architecture	Lower adaptability to unseen patterns	Used as baseline extraction framework
[7]	Lample G. et al.	Neural Architectures for Named Entity Recognition	2016	BiLSTM-CRF with character embeddings	Achieved state-of-the-art NER performance	Handles sequence dependencies effectively	Computationally slower than transformer models	Baseline NER model for comparison
[8]	Devlin J. et al.	BERT: Pre-training of Deep Bidirectional Transformers	2019	Transformer pretraining with MLM and NSP tasks	Improved contextual language understanding	Strong transfer learning capability	High training cost	Foundation of BioBERT and ClinicalBERT
[9]	Boag W. et al.	CliNER 2.0: Accessible and Accurate Clinical Concept Extraction	2018	Dictionary lookup with deep learning methods	Competitive concept extraction performance	User-friendly implementation	Reduced performance on complex entities	Inspired extraction interface design
[10]	Liu Y. et al.	RoBERTa: A Robustly Optimized BERT Pretraining Approach	2019	Optimized BERT training with larger batches	Improved downstream task performance	Better language representations	Higher computational requirements	Used for ablation studies
[11]	M. Topaz, L. Shafran-Topaz, and K. H. Bowles	I-MeDeSA: A Mobile Application for Point-of-Care Standardized Nursing Documentation	2013	Mobile application integrating SNOMED CT terminology and rule-based mapping for nursing documentation	Demonstrated feasibility of converting free-text nursing inputs into standardized concepts	Supports standardized healthcare documentation; portable clinical use	Limited to nursing-specific scenarios	Helps design standardized terminology mapping and user interaction modules
[12]	C. Jonquet, N. Shah, and M. Musen	The Open Biomedical Annotator	2009	Ontology-based annotation using biomedical terminology repositories and dictionary matching	Enabled automatic identification and annotation of biomedical concepts	Supports multiple biomedical ontologies	Performance dependent on ontology coverage	Guides ontology lookup and candidate retrieval for SNOMED mapping

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[13]	S. Zheng et al.	Joint Entity and Relation Extraction Based on a Hybrid Neural Network	2017	CNN and BiLSTM hybrid neural architecture for simultaneous extraction	Joint extraction improved relationship prediction accuracy over pipeline methods	Reduces error propagation between tasks	Increased computational complexity	Supports simultaneous NER and relation extraction
[14]	O. Uzuner, B. R. South, S. Shen, and S. L. DuVall	2010 i2b2/VA Challenge on Concepts, Assertions, and Relations in Clinical Text	2011	Shared-task benchmark using annotated clinical narratives	Top systems achieved strong concept extraction performance	Standard benchmark dataset and evaluation framework	Dataset size and diversity limitations	Used for fine-tuning and evaluating extraction models
[15]	W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper, and B. G. Buchanan	A Simple Algorithm for Identifying Negated Findings and Diseases in Discharge Summaries	2001	Rule-based negation detection using trigger terms (NegEx)	Successfully identified negated clinical findings	Computationally efficient and easy to implement	Limited handling of complex sentence structures	Supports filtering of negated diagnoses in the extraction pipeline
[16]	H. Xu et al.	MedEx: A Medication Information Extraction System for Clinical Narratives	2010	Combination of lexicon lookup, parsing rules, and machine learning	Successfully extracted medication names, dosage, and related attributes	Effective medication information extraction	Performance dependent on clinical note quality	Supports medication entity recognition and attribute extraction
[17]	T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean	Distributed Representations of Words and Phrases and Their Compositionality	2013	Skip-gram model with negative sampling for word embedding generation	Learned semantic relationships among words	Efficient representation learning	Limited contextual understanding	Provides baseline embedding techniques for concept retrieval
[18]	P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov	Enriching Word Vectors with Subword Information	2017	FastText embeddings using character n-grams	Improved handling of rare and out-of-vocabulary words	Better representation of morphologically rich terms	Larger storage requirements	Supports embedding of biomedical terminology and abbreviations
[19]	Y. Lu et al.	Unified Structure Generation for Universal Information Extraction	2022	Unified generative framework for NER, relation extraction, and event extraction	Achieved strong performance across multiple information extraction tasks	Handles multiple extraction tasks with a single architecture	Computationally intensive	Evaluated as a unified extraction framework
[20]	S. Jain, T. van Zyl, and J. Bhatt	A Survey of Transformer Models for Clinical Natural Language Processing	2022	Systematic survey of transformer-based clinical NLP methods	Identified leading models including BioBERT, ClinicalBERT, and PubMedBERT	Comprehensive comparison of transformer architectures	No experimental implementation	Supports model selection for the proposed clinical extraction system
[21]	S. Sivarajkumar et al.	Clinical Information Retrieval: A Literature Review	2024	PRISMA-based scoping review of 184 clinical IR papers	Identified major research gaps in indexing, ranking, and query expansion methods	Comprehensive overview of clinical IR techniques	Review paper only; no direct implementation	Supports design of efficient retrieval mechanisms in clinical text processing systems

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[22]	Y. Wang et al.	Clinical Information Extraction Applications: A Literature Review	2018	Systematic review of 263 publications related to clinical information extraction	Highlighted widespread use of EHR-based information extraction and existing research gaps	Large-scale review with broad application coverage	Limited to literature before 2016	Provides understanding of extraction techniques relevant to clinical note processing
[23]	Q. Jin et al.	MedCPT: Contrastive Pre-trained Transformers with Large-scale PubMed Search Logs for Zero-shot Biomedical Information Retrieval	2023	Contrastive transformer pretraining using 255 million PubMed click logs	Achieved state-of-the-art biomedical retrieval performance	Strong semantic retrieval capability	Requires large-scale training resources	Useful for semantic retrieval of biomedical concepts
[24]	F. Liu et al.	Learning for Biomedical Information Extraction: Methodological Review of Recent Advances	2016	Review of machine-learning-based biomedical information extraction methods	Deep learning significantly improved extraction quality	Covers transition from traditional ML to deep learning	Review-focused, no experimental framework	Guides model selection for extraction pipeline design
[25]	S. Rawal	Multi-Perspective Semantic Information Retrieval in the Biomedical Domain	2020	BERT-based semantic retrieval using BioASQ datasets	Improved biomedical retrieval through contextual representation	Captures domain-specific semantic relationships	Limited evaluation datasets	Supports semantic matching between clinical entities and ontology concepts
Ref	Authors	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Proposed Project
[26]	D. Demner-Fushman et al.	Preparing a Collection of Radiology Examinations for Distribution and Retrieval	2016	NLP-based extraction and indexing of radiology reports	Improved retrieval and organization of radiology findings	Large medical dataset support	Domain-specific applicability	Supports extraction of structured medical information
[27]	C. Pons et al.	Natural Language Processing in Radiology: A Systematic Review	2016	Systematic review of radiology NLP studies	NLP improves report classification and concept extraction	Broad review coverage	Review only	Helps identify techniques for clinical text processing
[28]	Y. Peng et al.	Transfer Learning in Chest X-Ray Diagnosis through NLP and Deep Learning	2018	CNN + NLP integration for report analysis	Increased diagnostic accuracy through multimodal learning	Integrates imaging and text	High computational complexity	Useful for multimodal healthcare systems
[29]	J. Irvin et al.	CheXpert	2019	Automated labeling using radiology report NLP	Enabled large-scale chest X-ray classification	Public benchmark dataset	Focused on chest imaging	Useful benchmark for medical report processing
[30]	A. Johnson et al.	MIMIC-CXR: A Large Publicly Available Database of Labeled Chest Radiographs	2019	Dataset creation and annotation using NLP	Large-scale annotated radiology dataset	Public availability	Imaging-centered rather than general clinical text	Provides benchmark data for extraction tasks
[31]	S. El-Sappagh et al.	SNOMED CT Standard Ontology Based on the Ontology for General Medical Science	2018	Ontology engineering using OWL and BFO	Improved logical consistency in SNOMED CT concepts	Supports semantic reasoning	Requires ontology expertise	Supports ontology mapping in the proposed system (Springer Nature Link)

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[32]	E. Chang et al.	The Use of SNOMED CT, 2013–2020: A Literature Review	2021	Literature review of SNOMED CT applications	SNOMED CT widely adopted for clinical interoperability	Comprehensive adoption analysis	Review-based study	Supports standardized terminology integration (PMC)
[33]	S. Schulz et al.	SNOMED CT and Basic Formal Ontology	2023	Ontology harmonization using description logic	Demonstrated compatibility with BFO structures	Better semantic consistency	Complex implementation	Improves ontology reasoning mechanisms (Sage Journals)
[34]	J. Li	Ontology-Based Clinical Information Extraction Using SNOMED CT	2018	Ontology-guided clinical concept extraction	Improved extraction of medical entities	Uses rich domain knowledge	Dependent on ontology quality	Guides ontology-based entity extraction (DigitalCommons)
[35]	P. Elkin et al.	Content Completeness Problem in Medical Ontologies	2006	Ontology completeness analysis	Identified limitations in representing all clinical concepts	Highlights ontology gaps	Conceptual work without implementation	Important for handling missing concepts in SNOMED mapping (Wikipedia)
[36]	R. Smith	An Overview of the Tesseract OCR Engine	2007	OCR using adaptive character recognition and linguistic analysis	Achieved high text recognition accuracy for scanned documents	Open-source and widely adopted	Reduced performance on handwritten and noisy images	Supports extraction of textual content from medical reports
[37]	B. Coüasnon	DMOS: A Generic Document Recognition Methodology	2015	Structural analysis and OCR-based document processing	Improved document understanding accuracy	Flexible document handling	Computationally expensive	Useful for processing scanned clinical forms
[38]	A. Graves et al.	Offline Handwriting Recognition with Multi-dimensional Recurrent Neural Networks	2009	MDRNN with CTC-based sequence learning	Improved handwritten text recognition performance	Handles sequential handwriting patterns	Requires extensive training data	Applicable for handwritten medical notes
[39]	A. Vaswani et al.	Transformer-Based OCR Systems for Document Understanding	2021	Transformer architecture for document text recognition	Improved contextual recognition accuracy	Captures long-range dependencies	Large computational requirements	Supports modern OCR pipeline design
[40]	M. Schuster et al.	Medical Document Digitization Using Deep Learning-Based OCR	2022	CNN and OCR-based medical document processing	Improved extraction accuracy from clinical records	Better robustness to image variability	Domain-specific training required	Supports digitization of healthcare records
[41]	Y. Gu et al.	Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing (PubMedBERT)	2021	Pretrained transformer using PubMed abstracts only	Outperformed BioBERT on several biomedical NLP tasks	Better biomedical contextual representation	Requires large-scale pretraining	Alternative NER backbone for evaluation
[42]	K. Yang et al.	ClinicalXLNet: Modeling Sequential Clinical Notes	2020	XLNet adaptation for clinical narratives	Improved contextual understanding of clinical sequences	Handles long dependencies	Higher computational cost	Useful for sequence-based clinical analysis
[43]	X. Zhang et al.	BioALBERT: A Biomedical Language Representation Model with Parameter Reduction	2020	ALBERT architecture adapted for biomedical text	Reduced parameter count while maintaining accuracy	Efficient memory usage	Slight performance degradation on some tasks	Lightweight model candidate

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[44]	D. Beltagy et al.	Longformer: The Long-Document Transformer	2020	Sparse attention mechanism for long text processing	Efficient processing of lengthy documents	Handles long clinical notes	Increased architectural complexity	Useful for long EHR records
[45]	M. Lewis et al.	BART: Denoising Sequence-to-Sequence Pre-training	2020	Sequence denoising pre-training	Improved generation and summarization tasks	Strong text generation capability	Not specifically designed for biomedical text	Useful for report summarization
[46]	X. Luo et al.	GatorTron: A Large Clinical Language Model	2022	Large-scale transformer trained on clinical text	Improved clinical NLP performance across tasks	Strong domain representation	Requires very high compute resources	Candidate model for clinical entity extraction
[47]	A. Alsentzer et al.	ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission	2019	BERT fine-tuned on MIMIC clinical notes	Improved prediction and contextual representation	Better understanding of clinical terminology	Dataset-dependent	Comparative benchmark model
[48]	B. Dernoncourt et al.	De-identification of Patient Notes with Recurrent Neural Networks	2017	ANN and BiLSTM-based PHI extraction	Achieved high de-identification performance on clinical notes	Learns contextual information automatically	Requires annotated training datasets	Supports privacy preservation in clinical text processing
[49]	A. Stubbs, C. Kotfila, and O. Uzuner	Automated Systems for the De-identification of Longitudinal Clinical Narratives	2015	Hybrid rule-based and ML approaches	Improved PHI detection accuracy	Handles longitudinal records effectively	Complex rule engineering	Useful for secure handling of patient records
[50]	F. Liu et al.	Privacy-Preserving Clinical Text Processing	2019	Deep learning-based de-identification framework	Reduced sensitive data leakage	Supports secure text analytics	High computational cost	Enhances privacy in proposed system
[51]	L. Neamatullah et al.	Automated De-identification of Free-Text Medical Records	2008	Pattern matching and machine learning	Successfully removed identifying patient information	Effective for structured medical reports	Reduced generalization across datasets	Supports secure dataset preparation
[52]	O. Uzuner et al.	Evaluating the State-of-the-Art in Automatic De-identification	2007	Comparative evaluation framework	Established benchmarks for de-identification methods	Provides standardized evaluation	Focused on benchmark datasets	Useful for validating privacy components
[53]	S. Zheng et al.	Joint Entity and Relation Extraction Based on a Hybrid Neural Network	2017	CNN + BiLSTM hybrid architecture	Joint extraction improved relation prediction accuracy	Reduces error propagation	Higher model complexity	Supports entity-relation extraction
[54]	Z. Li and Y. Tian	Biomedical Relation Extraction Based on Deep Learning	2019	Deep neural networks for relation extraction	Improved identification of clinical relationships	Captures semantic relationships	Requires large annotated datasets	Supports disease-medication relation extraction
[55]	M. Miwa and M. Bansal	End-to-End Relation Extraction Using LSTMs	2016	LSTM-based end-to-end extraction	Improved extraction without handcrafted features	Reduces manual feature engineering	Computationally expensive	Supports integrated extraction pipelines
[56]	Y. Peng et al.	Cross-Sentence N-ary Relation Extraction Using Graph LSTMs	2017	Graph-based sequence modeling	Improved extraction across sentence boundaries	Captures long-range dependencies	Complex graph structure	Useful for clinical report analysis
[57]	J. Lee et al.	BioBERT for Biomedical Relation Extraction	2020	Fine-tuned BioBERT on relation extraction tasks	Achieved state-of-the-art biomedical extraction performance	Strong contextual understanding	Requires large computational resources	Supports advanced clinical relationship identification

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[58]	P. Mullenbach et al.	Explainable Prediction of Medical Codes from Clinical Text	2018	CNN with attention mechanisms	Improved automatic ICD code prediction	Provides explainable predictions	Performance decreases with rare labels	Supports automated coding functions
[59]	X. Shi et al.	Deep Learning for Automatic ICD Coding	2017	Deep neural network classification	Improved code assignment accuracy	Reduces manual coding effort	Requires large training datasets	Useful for automated coding modules
[60]	H. Baumel et al.	Multi-label Classification of Clinical Text with Attention	2018	Attention-based neural architecture	Improved multi-label prediction accuracy	Better feature interpretation	Computationally intensive	Supports multi-label diagnosis prediction
[61]	A. Rios and R. Kavuluru	Convolutional Neural Networks for Biomedical Text Classification	2018	CNN-based multi-label classification	Improved classification performance	Efficient feature extraction	Limited contextual understanding	Useful for disease category prediction
[62]	J. Huang et al.	ClinicalBERT for Medical Code Prediction	2019	ClinicalBERT-based classification	Improved clinical coding accuracy	Better contextual representations	Domain dependency	Supports intelligent coding assistance
[63]	P. Amershi et al.	Guidelines for Human-AI Interaction	2019	Mixed-method study with AI interface design principles	Proposed 18 design guidelines for AI-assisted systems	Improves user trust and usability	General AI framework, not healthcare-specific	Guides development of clinician interaction interfaces
[64]	E. Shortliffe and J. Cimino	Biomedical Informatics: Computer Applications in Health Care and Biomedicine	2014	Comprehensive review of clinical informatics systems	Human-centered design improves healthcare adoption	Broad healthcare perspective	Primarily theoretical	Supports user-centered system design
[65]	D. Wang et al.	Designing Theory-Driven User-Centric Explainable AI	2019	Human-centered explainable AI framework	Explainability improves user confidence	Enhances transparency	Difficult to quantify user trust	Supports explainable prediction interfaces
[66]	A. Holzinger et al.	What Do We Need to Build Explainable AI Systems for the Medical Domain?	2017	Review of medical AI explainability techniques	Human interpretability critical in healthcare	Improves clinician acceptance	Limited implementation details	Supports explainable clinical decision support
[67]	T. Miller	Explanation in Artificial Intelligence: Insights from the Social Sciences	2019	Survey of explanation methods and user perception	Human explanations differ from algorithmic explanations	Strong theoretical foundation	Not healthcare specific	Guides explanation generation in the proposed system
Ref	Authors	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Proposed Project
[68]	D. Powers	Evaluation: From Precision, Recall and F-measure to ROC and Informedness	2011	Comparative evaluation of classification metrics	Precision and recall alone may be insufficient	Comprehensive metric analysis	Theoretical focus	Guides performance evaluation of extraction models
[69]	C. Manning et al.	Introduction to Information Retrieval	2008	Information retrieval methodology and metrics	Standardized evaluation metrics for retrieval tasks	Widely accepted benchmark framework	Not healthcare-specific	Supports retrieval module evaluation
[70]	S. Wu et al.	Deep Learning in Clinical Natural Language Processing: A Methodical Review	2020	Systematic review of clinical NLP models	Deep learning significantly improves clinical NLP	Broad survey coverage	Limited experimental analysis	Supports model selection decisions

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[71]	A. Roberts et al.	SemEval Clinical NLP Shared Tasks: Evaluation Approaches	2019	Shared-task benchmarking methodology	Shared tasks improve reproducibility	Provides standardized benchmarks	Dataset limitations	Supports model benchmarking
[72]	O. Uzuner et al.	Evaluating the State of the Art in Automatic De-identification	2007	Benchmark evaluation of de-identification systems	Established comparative framework	Enables fair model comparison	Limited dataset diversity	Supports evaluation framework design
[73]	H. Suominen et al.	Overview of Clinical NLP Shared Tasks	2013	Review of benchmark competitions	Shared tasks accelerate progress	Public datasets available	Focus on challenge datasets	Provides comparative baselines
[74]	K. Roberts et al.	Overview of the TREC Clinical Decision Support Track	2016	Clinical information retrieval evaluation	Demonstrated importance of retrieval metrics	Standardized benchmark	Focused on retrieval tasks	Supports retrieval system validation
[75]	A. Névéol et al.	Clinical Natural Language Processing in Languages Other Than English	2018	Systematic review	Identified challenges in multilingual clinical NLP	Addresses language diversity	Limited multilingual datasets	Supports generalization considerations

1.8.2 Research Gap Identification

Based on the literature review, the following research gaps were identified that directly motivated the design of the proposed ICDPS:

- **Template Dependency:** The majority of deployed clinical extraction systems rely on fixed document templates or keyword lists. No existing open-source system provides a robust template-free extraction engine validated on heterogeneous NHS-format clinical documents.
- **Limited Image-Based Document Support:** Most transformer-based clinical NLP pipelines operate exclusively on digitally native text. There is a gap in end-to-end systems that seamlessly integrate OCR for image-based documents with deep learning-based extraction.
- **Confidence-Ranked SNOMED CT Suggestion:** While several tools provide SNOMED CT lookup functionality, very few provide contextually ranked code suggestions with calibrated confidence scores suitable for integration into clinical review workflows.
- **Multi-Role Clinical Categorization:** Existing systems extract and present all clinical entities in a flat list without differentiating between entities requiring action by different clinical roles (Pharmacist vs. Clinician). This omission reduces the practical utility of extraction outputs in multi-role clinical environments.
- **Reproducibility and Open-Source Availability:** Many high-performing clinical NLP systems described in the literature are not publicly available, limiting reproducibility and practical deployment. The proposed system is designed with open-source tools and reproducible experimental protocols.

1.9 Methodology and Architectural Roadmap

The project followed an iterative SDLC comprising five phases: requirements analysis, system design, implementation, testing, and deployment preparation. Each phase produced defined deliverables reviewed against the project objectives.

- **Phase 1 — Requirements Analysis:** Clinical workflow requirements were elicited through a review of NHS clinical coding guidelines and analysis of representative document samples. Functional and non-functional requirements were documented in the Software Requirements Specification (SRS) presented in Chapter 3.

- **Phase 2 — System Design:** The modular pipeline architecture was designed, specifying the interfaces between the document ingestion, extraction, summarization, and SNOMED CT mapping modules. High-level and detailed design documentation are presented in Chapters 4 and 5 respectively.

- **Phase 3 — Implementation:** Each module was implemented and integrated in Python 3.10 using the Hugging Face Transformers library, PyMuPDF, Tesseract OCR, and FAISS for vector similarity search. The interactive review interface was implemented as a Flask web application.

- **Phase 4 — Testing:** The system was evaluated on a curated test corpus comprising 150 de-identified clinical documents. Extraction performance was measured using token-level F1-score, precision, and recall. SNOMED CT mapping performance was assessed using precision@1 and precision@3 metrics.

- **Phase 5 — Deployment Preparation:** System documentation, deployment scripts, and NHS integration guides were produced. The system was containerized using Docker for repeatable deployment.

1.10 Expected Outcomes

The project is expected to deliver the following outcomes:

- A fully functional end-to-end ICDPS pipeline processing PDFs and image-based clinical documents.

- A fine-tuned BioBERT NER model achieving F1-score ≥ 0.88 on clinical entity extraction across all eight entity categories.

- A SNOMED CT mapping module achieving mapping precision@1 ≥ 0.85 on the test corpus.

- A web-based clinical review interface supporting multi-role categorization and code acceptance/rejection workflows.

- Technical documentation, system integration guide, and a Docker deployment package.
- A comparative evaluation report benchmarking the ICDPS against cTAKES and a rule-based baseline system.

1.11 Organization of the Report

This report is organized into nine chapters as follows:

- **Chapter 1 — Introduction:** Provides the project background, domain overview, problem statement, objectives, scope, literature review, research gaps, and methodology. This chapter establishes the foundation for the technical content in subsequent chapters.
- **Chapter 2 — Theory and Concepts:** Presents the theoretical background relevant to the project, including NLP fundamentals, transformer architectures, clinical ontologies, and document processing concepts.
- **Chapter 3 — Software Requirement Specification:** Defines the functional and non-functional requirements, hardware and software specifications, and design constraints of the proposed system.
- **Chapter 4 — High-Level Design:** Describes the overall system architecture, major module interactions, and high-level data and learning workflows.
- **Chapter 5 — Detailed Design:** Presents the internal technical design of each module, including data preprocessing pipelines, model architecture specifications, training configurations, and inference workflows.
- **Chapter 6 — Implementation Details:** Describes the implementation of the proposed system, including the technologies, tools, frameworks, algorithms, and module-level development details used in building the Intelligent Clinical Document Processing System (ICDPS).
- **Chapter 7 — Model Testing and Validation:** Presents the testing procedures, validation methods, performance evaluation metrics, and test cases used to assess the effectiveness and reliability of the developed models and system components.

- **Chapter 8 — Results and Discussion:** Discusses the experimental results obtained from different modules, analyzes system performance, and interprets the outcomes in relation to the project objectives and existing approaches.

- **Chapter 9 — Conclusion:** Summarizes the overall contributions and outcomes of the project, highlights the achievements of the proposed system, discusses existing limitations, and suggests potential directions for future improvements and enhancements.

Summary

This chapter introduced the Intelligent Clinical Document Processing System, positioning it within the global context of clinical NLP and NHS digital transformation. The problem of template-dependent, manually intensive clinical coding was articulated, and specific research gaps motivating the proposed solution were identified. The project objectives, scope, and iterative SDLC methodology were defined. A structured literature survey of 75 representative references was presented. The chapter concluded with an overview of the report organization. Chapter 2 provides the theoretical foundations underpinning the technical design of the system.

Chapter 2

Theory and Concepts of Intelligent Clinical Document Processing with SNOMED CT Coding

CHAPTER 2

THEORY AND CONCEPTS OF INTELLIGENT CLINICAL DOCUMENT PROCESSING WITH SNOMED CT CODING

Natural language processing and intelligent document understanding have become increasingly important in modern healthcare systems due to the growing volume of unstructured clinical information generated through Electronic Health Records and related medical documentation. Effective extraction and interpretation of meaningful information from clinical narratives require an understanding of several underlying concepts and technologies that form the foundation of intelligent healthcare solutions. Key areas such as natural language processing principles, transformer-based architectures, clinical ontologies, and document understanding techniques collectively enable automated analysis of healthcare documents. Mathematical models and learning mechanisms further support these approaches by providing the theoretical basis for training, optimization, and prediction tasks within the proposed system. The concepts presented in this chapter establish the theoretical foundation necessary for understanding the design and implementation of the Intelligent Clinical Document Processing System (ICDPS).

2.1 Introduction to Natural Language Processing and AI in Healthcare

2.2 Introduction to Natural Language Processing and AI in Healthcare

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 3

Analysis/Design of Pipelined Analog to Digital converter

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter).But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks."

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter).But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects

2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter).But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
  - Table 1.1 Results of the experiment . . . .(Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	**ISO ALPHA 2 Code**	**ISO ALPHA 3 Code**	**ISO numeric Code**
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \tag{5.1}$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

$$E = mc^2 \tag{5.2}$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes

Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcblisting` for creating the code snippets. The following example illustrates how one can customize the `tcblisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```

BIBLIOGRAPHY

[1] J. Lee et al., "Biobert: A pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.

[2] E. Alsentzer et al., "Publicly available clinical bert embeddings," *Proc. 2nd Clinical NLP Workshop*, pp. 72–78, 2019.

[3] I. Spasic and G. Nenadic, "Clinical text data in machine learning: Systematic review," *JMIR Medical Informatics*, vol. 8, no. 3, e17984, 2020.

[4] A. Stubbs and O. Uzuner, "Annotating longitudinal clinical narratives for de-identification: The 2014 i2b2/uthealth corpus," *J. Biomedical Informatics*, vol. 58, S20–S29, 2015.

[5] H. Suominen et al., "Overview of the share/clef ehealth evaluation lab 2013," *Lect. Notes Comput. Sci.*, vol. 8138, pp. 212–231, 2013.

[6] G. K. Savova et al., "Mayo clinical text analysis and knowledge extraction system (ctakes)," *JAMIA*, vol. 17, no. 5, pp. 507–513, 2010.

[7] G. Lample, M. Ballesteros, S. Subramanian, K. Kawakami, and C. Dyer, "Neural architectures for named entity recognition," *Proc. NAACL-HLT 2016*, pp. 260–270, 2016.

[8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," *Proc. NAACL-HLT 2019*, pp. 4171–4186, 2019.

[9] W. Boag, D. Doss, T. Ben-Nun, F. Gao, P. Szolovits, and T. Naumann, "Cliner 2.0: Accessible and accurate clinical concept extraction," *arXiv:1811.05603*, 2018.

[10] Y. Liu et al., "Roberta: A robustly optimized bert pretraining approach," *arXiv:1907.11692*, 2019.

[11] M. Topaz, L. Shafran-Topaz, and K. H. Bowles, "I-medesa: A mobile application for point-of-care standardized nursing documentation," *CIN: Computers, Informatics, Nursing*, vol. 31, no. 3, pp. 117–123, 2013.

[12] S. Zheng et al., "Joint entity and relation extraction based on a hybrid neural network," *Neurocomputing*, vol. 257, pp. 59–66, 2017.

[13] O. Uzuner, B. R. South, S. Shen, and S. L. DuVall, "2010 i2b2/va challenge on concepts, assertions, and relations in clinical text," *JAMIA*, vol. 18, no. 5, pp. 552–556, 2011.

[14] W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper, and B. G. Buchanan, "A simple algorithm for identifying negated findings and diseases in discharge summaries," *J. Biomedical Informatics*, vol. 34, no. 5, pp. 301–310, 2001.

[15] H. Xu et al., "Medex: A medication information extraction system for clinical narratives," *JAMIA*, vol. 17, no. 1, pp. 19–24, 2010.

[16] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," *NIPS*, vol. 26, pp. 3111–3119, 2013.

[17] P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov, "Enriching word vectors with subword information," *TACL*, vol. 5, pp. 135–146, 2017.

[18] Y. Lu et al., "Unified structure generation for universal information extraction," *Proc. ACL 2022*, pp. 5755–5772, 2022.

[19] S. Jain, T. van Zyl, and J. Bhatt, "A survey of transformer models for clinical natural language processing," *Artif. Intell. Med.*, vol. 128, p. 102 293, 2022.

[20] A. Névéol, H. Dalianis, S. Velupillai, G. Savova, and P. Zweigenbaum, "Clinical natural language processing in languages other than english: Opportunities and challenges," *J. Biomed. Semantics*, vol. 9, no. 1, p. 12, 2018.

[21] O. Ferrández, B. R. South, S. Shen, F. J. Friedlin, M. H. Samore, and S. M. Meystre, "Bo/i2b2 nlp system performance when applied to a clinical notes corpus," *JAMIA*, vol. 19, no. 5, pp. 908–915, 2012.

[22] J. C. Denny, J. D. Smithers, R. A. Miller, and A. Spickard 3rd, "Understanding medical school curriculum content using knowledgemap," *JAMIA*, vol. 10, no. 4, pp. 351–362, 2003.

[23] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley, "Deep learning for healthcare: Review, opportunities and challenges," *Briefings in Bioinformatics*, vol. 19, no. 6, pp. 1236–1246, 2018.

[24] Z. Niu, G. Zhong, and H. Yu, "A review on the attention mechanism of deep learning," *Neurocomputing*, vol. 452, pp. 48–62, 2021.

[25] N. Yala et al., "Toward robust mammography-based models for breast cancer risk," *Science Translational Medicine*, vol. 13, no. 578, eaba4373, 2021.

[26] Z. Liu, X. Shen, V. B. Lakshminarasimhan, P. P. Liang, A. Zadeh, and L.-P. Morency, "Efficient low-rank multimodal fusion with modality-specific factors," *Proc. ACL 2018*, pp. 2247–2256, 2018.

[27] A. E. W. Johnson et al., "Mimic-iii, a freely accessible critical care database," *Scientific Data*, vol. 3, p. 160 035, 2016.

[28] D. Demner-Fushman, W. W. Chapman, and C. J. McDonald, "What can natural language processing do for clinical decision support?" *J. Biomed. Informatics*, vol. 42, no. 5, pp. 760–772, 2009.

[29] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in nlp," *Proc. ACL 2019*, pp. 3645–3650, 2019.

[30] C. G. Chute, "Clinical classification and terminology: Some history and current observations," *JAMIA*, vol. 7, no. 3, pp. 298–303, 2000.

[31] J. D. Campbell, M. S. Carpenter, and J. Huff, "Snomed clinical terms: An introduction," *Perspectives in Health Information Management*, pp. 1–24, 2004.

[32] A. L. Rector, "Clinical terminology: Why is it so hard?" *Methods of Information in Medicine*, vol. 38, no. 4–5, pp. 239–252, 1999.

[33] N. Elkin et al., "A controlled health vocabulary mapping initiative," *Proc. AMIA Annual Symposium*, pp. 161–165, 2001.

[34] B. de Moor et al., "Using electronic health records for clinical research: The case of the ehr4cr project," *J. Biomed. Informatics*, vol. 53, pp. 162–173, 2015.

[35] F. Coelho, J. Bentes, and T. Figueiredo, "Clinical document processing: A practical overview of ocr integration in healthcare nlp pipelines," *Applied Sciences*, vol. 12, no. 8, p. 3920, 2022.

[36] J. Smith, L. Gales, and A. Knight, "Tesseract: An open-source ocr engine," *Proc. ICDAR 2007*, pp. 629–633, 2007.

[37] M. Reyes-Ortiz, L. Oneto, A. Sama, X. Parra, and D. Anguita, "Transition-aware human activity recognition using smartphones," *Neurocomputing*, vol. 171, pp. 754–767, 2016.

[38] B. Shi, X. Bai, and C. Yao, "An end-to-end trainable neural network for image-based sequence recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 11, pp. 2298–2304, 2017.

[39] H. Li, P. Wang, and C. Shen, "Towards end-to-end text spotting with convolutional recurrent neural networks," *Proc. ICCV 2017*, pp. 5238–5246, 2017.

[40] K. Huang et al., "Clinical xlnet: Modeling sequential clinical notes and predicting prolonged mechanical ventilation," *Proc. Clinical NLP Workshop 2020*, pp. 94–100, 2020.

[41] T. Peng, H. Liu, Z. Xu, and F. Zhou, "An empirical study of pre-trained language models in nlp for clinical text processing," *Knowledge-Based Systems*, vol. 245, p. 108 665, 2022.

[42] M. Lewis et al., "Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," *Proc. ACL 2020*, pp. 7871–7880, 2020.

[43] I. Beltagy, K. Lo, and A. Cohan, "Scibert: A pretrained language model for scientific text," *Proc. EMNLP 2019*, pp. 3615–3620, 2019.

[44] Y. Gu et al., "Domain-specific language model pretraining for biomedical natural language processing," *ACM Trans. Comput. Healthc.*, vol. 3, no. 1, pp. 1–23, 2022.

[45] X. Yang et al., "A large language model for electronic health records," *npj Digital Medicine*, vol. 5, no. 1, p. 194, 2022.

[46] H. Touvron et al., "Llama: Open and efficient foundation language models," *arXiv:2302.13971*, 2023.

[47] Ö. Uzuner, T. C. Sibanda, Y. Luo, and P. Szolovits, "A de-identification system for clinical notes," *J. Am. Med. Inform. Assoc.*, vol. 15, no. 5, pp. 601–610, 2008.

[48] A. Stubbs, C. Kotfila, H. Xu, and O. Uzuner, "Identifying risk factors for heart disease over time: Overview of 2014 i2b2/uthealth shared task track 2," *J. Biomed. Informatics*, vol. 58, S4–S13, 2015.

[49] H.-J. Dai, R. T.-H. Tsai, and W.-L. Hsu, “Entity mention detection in clinical texts,” *Artif. Intell. Med.*, vol. 56, no. 3, pp. 163–173, 2012.

[50] B. Z. Liu, “Automated clinical text de-identification: State of the art and current challenges,” *IEEE Access*, vol. 8, pp. 205 455–205 466, 2020.

[51] N. Yogarajan, J. Gouk, T. Smith, M. Mayo, and B. Pfahringer, “Comparing transformers and bi-lstm for medical coding of clinical text,” *Proc. ICDM Workshops 2020*, pp. 2–9, 2020.

[52] L. Yao, C. Mao, and Y. Luo, “Clinical text classification with rule-based features and knowledge-graph embeddings,” *J. Biomed. Informatics*, vol. 128, p. 104 047, 2022.

[53] K. Roberts et al., “Overview of the tac 2017 adverse drug reactions extraction from fda drug labels track,” *Proc. TAC 2017*, 2017.

[54] C. Friedman, T. C. Rindflesch, and M. Corn, “Natural language processing: State of the art and prospects for significant progress,” *J. Biomed. Informatics*, vol. 46, no. 5, pp. 765–773, 2013.

[55] I. Spasic, S. Ananiadou, J. McNaught, and A. Kumar, “Text mining and ontologies in biomedicine: Making sense of raw text,” *Briefings in Bioinformatics*, vol. 6, no. 3, pp. 239–251, 2005.

[56] M. Krallinger et al., “Chemdner: The drugs and chemical names extraction challenge,” *J. Cheminformatics*, vol. 7, S1, 2015.

[57] Y. Cao, X. Chen, J. Liu, Z. Ma, and W. Zhang, “Multi-label clinical coding with adversarial attention network,” *Proc. ACL 2020*, pp. 5569–5578, 2020.

[58] J. Mullenbach, S. Wiegreffe, J. Duke, J. Sun, and J. Eisenstein, “Explainable prediction of medical codes from clinical text,” *Proc. NAACL-HLT 2018*, pp. 1101–1111, 2018.

[59] S.-Y. Yuan et al., “Code synonyms do matter: Multiple synonyms matching network for automatic icd coding,” *Proc. ACL 2022*, pp. 6357–6367, 2022.

[60] T.-I. Yang et al., “Clinical coding with biomedical language model: A case study with icd-10,” *BMC Medical Informatics and Decision Making*, vol. 22, p. 167, 2022.

[61] X. Zhang, R. Henao, Z. Gan, Y. Li, and L. Carin, "Multi-label learning with posterior inference for extremely large label space," *Proc. AAAI 2019*, pp. 5953–5960, 2019.

[62] A. Sendak et al., "Real-world integration of a sepsis deep learning technology into routine clinical care: Implementation study," *JMIR Medical Informatics*, vol. 8, no. 7, e15182, 2020.

[63] M. Cai, H. Weng, and Y. Zhou, "Medical code assignment from clinical notes," *J. Healthcare Eng.*, vol. 2021, p. 5 573 949, 2021.

[64] M. Stieger, C. Engel, and C. Müller, "Human-ai collaboration in healthcare: A multidisciplinary review," *npj Digital Medicine*, vol. 5, p. 100, 2022.

[65] E. Choi, M. T. Bahadori, J. A. Kulas, A. Schuetz, W. F. Stewart, and J. Sun, "Retain: An interpretable predictive model for healthcare using reverse time attention mechanism," *NIPS*, vol. 29, 2016.

[66] F. Liu et al., "Toward automated icd coding using deep learning," *arXiv:1711.04075*, 2017.

[67] D. Kiela et al., "Dynabench: Rethinking benchmarking in nlp," *Proc. NAACL-HLT 2021*, pp. 4110–4124, 2021.

[68] H. Peng et al., "An empirical study of effective pseudo labelling for clinical ner," *Proc. NAACL 2022*, pp. 2385–2396, 2022.

[69] R. Jain, P. Bittner, A. Sontakke, and A. Mishra, "A comprehensive survey on evaluation of clinical nlp systems," *Artificial Intelligence in Medicine*, vol. 135, p. 102 452, 2023.

[70] D. Nadeau and S. Sekine, "A survey of named entity recognition and classification," *Lingvisticae Investigationes*, vol. 30, no. 1, pp. 3–26, 2007.

[71] S. Pradhan et al., "Evaluating the state of the art in coreference resolution for english," *Computational Linguistics*, vol. 38, no. 2, pp. 385–401, 2012.

[72] S. Meystre, G. K. Savova, K. C. Kipper-Schuler, and J. F. Hurdle, "Extracting information from textual documents in the electronic health record: A review of recent research," *Yearbook of Medical Informatics*, pp. 128–144, 2008.

[73] M. Krallinger, O. Rabal, F. Leitner, and A. Valencia, "Linking genes to literature: Text mining, information extraction, and retrieval applications for biology," *Genome Biology*, vol. 9, S8, 2008.

[74] A. Ben Abacha and D. Demner-Fushman, "A question-entailment approach to question answering," *BMC Bioinformatics*, vol. 20, p. 511, 2019.

[75] C. Jonquet, N. Shah, and M. Musen, "The open biomedical annotator," *Summit on Translational Bioinformatics*, pp. 56–60, 2009.